Software Requirements Specification

for

Placement Preparing Assistant

Version 1.0 approved

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Revision History

Name	***Date***	***Reason For Changes***	***Version***

1. Introduction

1.1 Purpose

The placement season for students in their third and fourth year is very hectic. A lot of companies show up to our campus, looking to recruit some of the best talents in our college. Students apply for a lot of these companies, but they lack the structured company-specific preparation guidance in the short span of time. They end up preparing for topics that might not even be important for the interviews or Oral Assessment for that company. This results in suboptimal interview performance.

1.2 Document Conventions

<Describe any standards or typographical conventions that were followed when writing this SRS, such as fonts or highlighting that have special significance. For example, state whether priorities for higher-level requirements are assumed to be inherited by detailed requirements, or whether every requirement statement is to have its own priority.>

1.3 Intended Audience and Reading Suggestions

This SRS is for our developers of our software, our teacher(s), the product managers and the users (other NITJ students).

1.4 Product Scope

The software provides assistance to students, preparing for multiple interviews for different companies and helping them secure their internship/job by allowing them to solely focus on preparing for the interview rather than researching on the type of questions asked in them. College can provide this as a resource for students to be well prepared for interviews to help them get the best packages from the campus.

1.5 References

<List any other documents or Web addresses to which this SRS refers. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. Provide enough information so that the reader could access a copy of each reference, including title, author, version number, date, and source or location.>

2. Overall Description

2.1 Product Perspective

The Placement Preparation Assistance System (PPAS) is a standalone, college-specific web-based application developed for students of Dr. B.R. Ambedkar National Institute of Technology, Jalandhar (NIT Jalandhar). It is a new, self-contained system that does not form part of any larger existing system.

The system is designed to address a well-identified gap in the current placement preparation ecosystem: while platforms like LeetCode, InterviewBit, and GeeksforGeeks offer general preparation material, none of them provide company-specific, college-contextualized guidance. PPAS fills this gap by combining historical placement data from NITJ, publicly available interview experiences, and AI-driven summarization to deliver targeted, actionable preparation strategies.

The system interacts with external services including an LLM API (for summarization and recommendation), public job and interview platforms (for data aggregation), and a relational database for persistent storage of company records, student profiles, and question banks.

2.2 Product Functions

The major functions of PPAS are summarized below:

Student-Facing Functions:

- *User registration, login, and profile management*
- *Browsing and filtering companies visiting NITJ for placements or internships*
- *Viewing company-specific preparation roadmaps including relevant topics, question types, interview rounds, and difficulty levels*
- *AI-generated summaries of job roles, required skills, and recommended preparation strategies*
- *Progress tracking to monitor topics covered and readiness level*
- *Personalized recommendations based on registered companies and academic profile*

Admin-Facing Functions:

- *Managing the company database including adding, updating, and removing company records*
- *Managing the question and topic database*
- *Viewing analytics on student engagement and preparation trends*
- *Controlling access and maintaining data quality*

System-Level Functions:

- *Aggregating and processing historical NITJ placement data*
- *Fetching and parsing publicly available interview experiences and OA question patterns*
- *Generating structured preparation summaries using LLM integration*
- *Storing and retrieving all data through a secure relational database.*

2.3 User Classes and Characteristics

1) ***Pre-Placement Students (Primary Users):*** *These are third and fourth year undergraduate or postgraduate students at NITJ who are actively participating in the placement or internship drive. They are the primary target audience. They are expected to be technically literate but may not be experienced with placement processes. They will interact with the system frequently during the placement season to browse companies, view preparation material, and track their progress. This group is expected to have the highest usage frequency and the broadest range of technical backgrounds.*
2) ***Veteran/Experienced Students (Secondary Users):*** *These are students who have prior internship or placement experience and may have a clearer sense of their preparation needs. They are less likely to rely on the system heavily but may use it selectively, for example to quickly check OA patterns or interview round structures for a specific company.*
3) ***System Administrators:*** *These are the maintainers of the system, likely faculty coordinators or student developers. They are responsible for keeping the company database accurate and up to date, managing question banks, and monitoring system health. They are expected to be technically proficient.*
4) ***Placement Cell / Training and Placement Office (TPO):*** *These are institutional stakeholders who may use the system to understand student preparation trends, identify gaps, and plan placement-related workshops or resources. They interact primarily with analytics dashboards and do not modify academic content directly.*

2.4 Operating Environment

Server Environment:

- *Operating System: Linux (Ubuntu 22.04 LTS or equivalent)*
- *Backend Framework: Flask (Python 3.10+)*
- *Database: MySQL 8.0 or equivalent relational database*
- *Hosting: Cloud-based (e.g., AWS, Azure, or GCP) or on-premise institutional server*
- *Network: HTTPS over TLS 1.2 or higher*

Client Environment:

- *Operating System: Windows, macOS, Linux, Android, or iOS*
- *Browser: Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari (latest two stable versions)*
- *Minimum RAM: 2 GB on client device*
- *Network: Stable internet connection required for LLM API calls and data retrieval*

External Dependencies:

a) *LLM API (e.g., OpenAI GPT or equivalent) for AI-based summarization and recommendations*
b) *Public interview experience platforms for data aggregation*
c) *SMTP server for future email notification support*

2.5 Design and Implementation Constraints

1) ***Limited Historical Data:*** *The availability of structured, digitized historical placement data for NITJ is limited. Much of the historical data may need to be manually curated or scraped from public sources, which affects the accuracy and completeness of recommendations in early versions.*
2) ***AI API Cost:*** *Usage of external LLM APIs incurs per-request costs. The system must be designed to minimize unnecessary API calls through caching and batched processing to keep operational costs manageable.*
3) ***Single Institution Scope:*** *Version 1.0 is scoped exclusively for NIT Jalandhar. The architecture should be modular enough to allow extension to other institutions in future versions, but no multi-tenancy is required now.*
4) ***No Mobile App in Version 1.0:*** *The system will be web-based and accessible via mobile browsers, but a dedicated native mobile application is out of scope for this version.*
5) ***Regulatory Constraints:*** *All student data must be handled in accordance with applicable data privacy norms. No personally identifiable information (PII) shall be shared with external APIs without anonymization.*
6) ***Technology Stack Constraint:*** *The project is constrained to use Flask (Python) for the backend and MySQL for the database, as per project guidelines.*
7) ***Internet Dependency:*** *The system requires an active internet connection for LLM API access and external data fetching. Offline functionality is not supported in Version 1.0.*

2.6 User Documentation

The following documentation shall be provided with the system:

1) ***User Manual:*** *A step-by-step guide for students covering registration, company browsing, viewing preparation material, and tracking progress.*
2) ***Admin Manual:*** *A guide for administrators covering company and question database management, and use of the analytics dashboard.*
3) ***Installation Guide:*** *Instructions for setting up the application on a local machine or server, including environment setup, dependency installation, and database configuration.*
4) ***README File:*** *A concise quick-start guide hosted in the project repository covering basic setup and usage.*
5) ***In-App Help Tooltips:*** *Contextual guidance embedded within the application interface to assist first-time users.*

Since Version 1.0 is not deployed at scale and will require installation on individual devices or a local server, the installation guide is particularly important and will be maintained in the project repository.

2.7 Assumptions and Dependencies

Assumptions:

1) *Students are assumed to have access to a device with a modern web browser and a stable internet connection.*

2) *The placement cell will provide or validate historical company visit data for NITJ to seed the initial database.*
3) *Interview experiences and OA question patterns sourced from public platforms are assumed to be sufficiently representative of actual interview processes.*
4) *Users are assumed to be honest in their profile setup, particularly regarding their year of study and registered companies.*
5) *The LLM API used for summarization is assumed to be available and reliable during peak placement season usage.*

Dependencies:

1) *The system depends on access to an active LLM API subscription for its core summarization and recommendation features. Loss of API access will degrade core functionality.*
2) *The system depends on MySQL 8.0 or compatible database software being correctly installed and configured.*
3) *Frontend functionality depends on Bootstrap 5 and standard HTML5/CSS3/JavaScript support in the user's browser.*
4) *Future*

3. External Interface Requirements

The Placement Preparation Assistance System (PPAS) shall provide a web-based graphical user interface (GUI) accessible through standard web browsers.

3.1 User Interfaces

The following user interfaces shall be implemented:

1) ***Student Interface***
    a) *Registration/Login Page*
    b) *Dashboard*
    c) *List of Companies*
    d) *Company Preparation Details Page*
    e) *Topic Recommendation View for each Company*
    f) *Progress Tracking View*
    g) *Profile Management Page*

2) ***Administrator Interface***
    a) *Admin Login Page*
    b) *Company Management Panel*
    c) *Question/Topic Database Management*
    d) *Analytics Dashboard*

3.2 Hardware Interfaces

The Placement Preparation Assistance System is a web-based application and does not directly interface with specialized hardware components.

3.2.1 Supported Hardware Platforms

The system shall support execution on:

- *Laptops*
- *Smartphones*

3.2.2 Client Hardware Requirements

Minimum client requirements:

- *Device capable of running a modern web browser*
- *Minimum 2GB RAM recommended*
- *Stable internet connection*

3.2.3 Server Hardware Requirements

The system shall be deployed on a cloud or on-premise server with:

- *Multi-core processor*
- *Minimum 2GB RAM*
- *Persistent storage for database operations*

No direct interaction with peripheral hardware (e.g., printers, biometric devices, scanners) is included in Version 1.0.

3.3 Software Interfaces

3.3.1 Operating System

Server-side:

- *Linux (Ubuntu 22.04 LTS or equivalent)*

Client-side:

- *Any OS capable of running modern browsers (Windows, macOS, Linux, Android, iOS)*

3.3.2 Web Browser Interface

Supported browsers:

- *Google Chrome (latest 2 stable versions)*

- *Mozilla Firefox (latest 2 stable versions)*
- *Microsoft Edge (latest 2 stable versions)*
- *Safari (latest 2 stable versions)*

3.3.3 Database Interface

The system shall interface with:

- *MySQL 8.0 (or equivalent relational database system)*

Data exchanged between application and database includes:

- *Student profiles*
- *Company records*
- *Interview rounds*
- *Question banks*
- *Preparation topics*
- *Progress tracking data*

Communication shall occur through secure database connectors using SQL queries.

3.3.4 Backend Framework

The system shall use:

- *Flask (Python 3.10+) for backend processing*

Services provided by backend include:

- *Authentication handling*
- *Session management*
- *Data retrieval and storage*
- *Recommendation logic execution*

3.3.5 External Libraries and Tools

- *HTML5, CSS3, JavaScript (Frontend)*
- *Bootstrap 5 (UI Framework)*
- *RESTful API architecture*
- *JSON for data exchange*

All inter-component communication shall use structured JSON payloads.

No global shared memory mechanisms shall be used. All data sharing shall occur through database transactions or API calls.

3.4 Communications Interfaces

3.4.1 Network Protocols

The system shall use:

- *HTTP/1.1 or HTTP/2 for communication*
- *HTTPS (TLS 1.2 or higher) for secure communication*

All sensitive data transmission shall occur over encrypted channels.

3.4.2 Client-Server Communication

- *Communication shall follow RESTful API principles.*
- *Data shall be exchanged in JSON format.*
- *Authentication shall use secure session-based or token-based mechanisms.*

3.4.3 Email Communication

If email notifications are implemented (future scope), the system shall:

- *Use SMTP protocol*
- *Support automated email notifications for:*
    - *Registration confirmation*
    - *Password reset*
    - *Placement updates*

3.4.4 Security Requirements

- *All user credentials shall be encrypted before storage.*
- *Passwords shall be hashed using industry-standard hashing algorithms.*
- *HTTPS shall be mandatory for login and data submission.*
- *Session tokens shall expire after a defined timeout period.*

3.4.5 Performance Requirements

- *System shall support concurrent access by at least 500 users.*
- *Average page load time shall not exceed 3 seconds under normal load.*
- *Data synchronization between client and server shall occur in real time upon user action.*

4. System Features

4.1 Company-Specific Preparation Roadmap Generation

4.1.1 Description and Priority

Priority: High. *When a student selects a company they have registered for, the system generates a structured preparation roadmap covering interview rounds, frequently tested topics, difficulty levels, and an AI-generated summary of what to focus on. This is the core feature of PPAS.*

4.1.2 Stimulus/Response Sequences

- *Student selects a company → System displays round-wise breakdown, topic list, and AI-generated preparation summary.*
- *Cached data is outdated → System triggers a fresh LLM API call and updates the summary.*
- *No historical NITJ data exists for a company → System falls back to a general role-based preparation guide.*

4.1.3 Functional Requirements

- ***REQ-1:*** *The system shall maintain a database of companies that have visited NITJ, including job role, package, year of visit, and round structure.*
- ***REQ-2:*** *The system shall display a round-wise breakdown of the interview process for each company.*
- ***REQ-3:*** *The system shall use an LLM API to generate a concise, actionable preparation summary based on aggregated data.*
- ***REQ-4:*** *LLM-generated summaries shall be cached and refreshed at a configurable interval or upon admin request.*
- ***REQ-5:*** *When no NITJ-specific data is available, the system shall fall back to a general role-based preparation guide.*

4.2 Student Progress Tracking

4.2.1 Description and Priority

Priority: Medium-High *Students can mark topics as completed, set preparation goals, and view a visual summary of their readiness across all companies they are registered for.*

4.2.2 Stimulus/Response Sequences

- *Student marks a topic as complete → System updates the database and recalculates completion percentage in real time.*
- *Student opens dashboard → System displays progress bars and completion status for all registered companies.*
- *Student deregisters from a company → System archives associated progress data and removes it from the active list.*

4.2.3 Functional Requirements

- ***REQ-1:*** *The system shall allow students to mark individual topics and tasks as complete or incomplete.*
- ***REQ-2:*** *The system shall calculate and display a preparation completion percentage per company.*
- ***REQ-3:*** *The dashboard shall show a consolidated progress view across all registered companies.*
- ***REQ-4:*** *Progress data shall persist in the database across sessions and logouts.*

4.3 Company and Question Database Management

4.3.1 Description and Priority

Priority: High*. Administrators can add, edit, and delete company records, interview round details, and question/topic entries through a dedicated admin panel to keep system data accurate and current.*

4.3.2 Stimulus/Response Sequences

- *Admin navigates to Company Management Panel → System displays all company records with options to add, edit, or delete.*
- *Admin submits a new company record → System validates input, stores the record, and makes it immediately available to students.*
- *Admin updates the question bank → System updates records and invalidates related cached LLM summaries.*

4.3.3 Functional Requirements

- ***REQ-1:*** *The system shall provide full CRUD functionality for company records via the admin interface.*
- ***REQ-2:*** *Each company record shall include company name, job roles, years of NITJ visits, package offered, round structure, and topic tags.*
- ***REQ-3:*** *The system shall provide an interface to add, edit, categorize, and delete question and topic bank entries.*
- ***REQ-4:*** *The system shall invalidate cached LLM summaries when underlying data is modified.*
- ***REQ-5:*** *The admin panel shall be accessible only to authenticated users with administrator privileges.*

4.4 AI-Powered Role Summarization

4.4.1 Description and Priority

Priority: High *The system provides AI-generated summaries of job roles, covering responsibilities, required skills, and how preparation differs across roles (e.g., SDE vs. Data Analyst). This helps students unfamiliar with the roles they are applying for.*

4.4.2 Stimulus/Response Sequences

- *Student selects a job role on a company page → System displays an AI-generated role summary with required skills and interview focus areas.*
- *Student browses the general role library → System lists all common NITJ placement roles with individual summaries.*

4.4.3 Functional Requirements

- ***REQ-1:*** *The system shall maintain a library of job roles commonly offered during NITJ placements.*
- ***REQ-2:*** *For each role, the system shall provide an AI-generated summary covering responsibilities, required skills, and interview focus areas.*
- ***REQ-3:*** *Role summaries shall be cached and linked from the company preparation page for contextual access.*

4.5 Analytics Dashboard

4.5.1 Description and Priority

Priority: Medium *An admin-facing dashboard providing insights into student engagement, most-viewed companies, frequently accessed topics, and overall preparation trends to help the placement cell identify gaps and plan resources.*

4.5.2 Stimulus/Response Sequences

- *Admin opens the Analytics Dashboard → System aggregates usage data and displays visualizations of company views, topic access frequency, and student progress distribution.*

4.5.3 Functional Requirements

- ***REQ-1:*** *The system shall log anonymized student interaction events including page views, topic completions, and session durations.*
- ***REQ-2:*** *The dashboard shall display aggregated metrics including top-viewed companies, most-accessed topics, and student progress distribution.*
- ***REQ-3:*** *Analytics shall be accessible only to users with administrator or placement cell privileges.*
- ***REQ-4:*** *No personally identifiable information shall be displayed; all metrics shall be aggregated or anonymized.*

5. Other Nonfunctional Requirements

5.1 Data Encryption and integrity

Student personal information must remain safe and confidential with the software in the database.

5.2 Safety Requirements

- *The system shall prevent accidental data deletion through confirmation dialogs.*
- *Administrative deletion of company records shall require explicit confirmation.*

- *The system shall maintain regular automated database backups.*
- *Recovery procedures shall be documented and tested quarterly.*

5.3 Security Requirements

5.3.1 Authentication

- *All users shall authenticate using a unique username/email and password.*
- *Passwords shall meet minimum complexity requirements:*
    - *Minimum 8 characters*
    - *At least one uppercase letter*
    - *At least one numeric digit*
- *The system shall lock an account after 5 consecutive failed login attempts.*
- *Password reset shall require verified email authentication.*

5.3.2 Authorization

The system shall implement role-based access control (RBAC):

- *Student users:*
    - *View company preparation material*
    - *Track personal progress*
    - *Update personal profile*
- *Admin users:*
    - *Add, update, or delete company data*
    - *Modify interview round information*
    - *Access analytics dashboard*

5.3.3 Privacy Protection

- *Student data shall not be shared with third parties.*
- *Only authorized administrators may access aggregated student analytics.*
- *The system shall comply with institutional data privacy policies.*

5.4 Software Quality Attributes

5.4.1 Availability

- *The system shall maintain at least **99% uptime** during placement season.*
- *Scheduled maintenance shall be announced 24 hours in advance.*

5.4.2 Performance

- *Average page response time shall not exceed 3 seconds under normal load (≤500 concurrent users).*
- *Database queries shall return results within 1 second under standard conditions.*

5.4.3 Reliability

- *System failure rate shall not exceed 1% of total user sessions.*
- *Data loss tolerance shall be zero for committed transactions.*

5.4.4 Usability

- *A new user shall be able to navigate to company preparation content within 3 clicks after login.*
- *The interface shall be intuitive and require no formal training.*
- *Error messages shall clearly describe corrective actions.*

5.4.5 Maintainability

- *Code shall follow modular architecture.*
- *Backend components shall be documented.*
- *All APIs shall include descriptive comments.*
- *System shall allow addition of new companies without modifying core architecture.*

5.4.6 Portability

- *The system shall be deployable on Linux-based servers.*
- *The frontend shall operate on major modern browsers without modification.*

5.4.7 Scalability

- *The system architecture shall support horizontal scaling via cloud deployment.*
- *Database indexing shall support growth up to 10,000+ student records without significant performance degradation.*

5.5 Business Rules

- *Only registered NITJ students may create accounts.*
- *Students may access only their own progress data.*
- *Only administrators may modify company preparation data.*
- *Company data must be approved by faculty before being published.*
- *Historical company data shall remain archived even after placement season ends.*
- *Student accounts shall be deactivated after graduation unless extended by admin approval.*
- *Analytics reports shall present anonymized data when viewed by faculty.*